Worksheet No: 3

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Subject Name: Advance Database Management System **Subject Code:** 24CAP-607

1. **Aim of the practical:** Write a query to perform relation employee.

2. **Objective:** Create a database named "relation" and define an employee table (EMP) with employee details such as employee number, name, and title.

Populate the EMP table with records, categorize employees into specific roles, and perform various queries to retrieve employee information based on title criteria.,

3. Commands/Steps:

Step-1: Created Database:
create database relation;
Query OK, 1 row affected (0.02 sec)

Step-2: Use this database: mysql>
use relation;
Database changed

Setp-3: Created table EMP:

mysql> create table EMP(Eno varchar(5), Ename varchar(20), Title varchar(20)); Query OK, 0 rows affected (0.07 sec)

Step-4: Inserted 8 records into the EMP table with 3 Attributes . insert into EMP values("E1", "J.Doe", "Elect.Eng");
Query OK, 1 row affected (0.02 sec)

mysql> insert into EMP values("E2", "M.Smith", "Syst.Anal");
Query OK, 1 row affected (0.01 sec)

mysql> insert into EMP values("E3", "A.Lee", "Mech.Eng");
Query OK, 1 row affected (0.01 sec)

mysql> insert into EMP values("E4", "J.Miller", "Programmer");
Query OK, 1 row affected (0.01 sec)

mysql> insert into EMP values("E5", "B.Casey", "Syst.Anal");
Query OK, 1 row affected (0.01 sec)

mysql> insert into EMP values("E6", "L.Chu", "Elect.Eng");
Query OK, 1 row affected (0.01 sec)

mysql> insert into EMP values("E7", "R.Davis", "Mech.Eng");
Query OK, 1 row affected (0.01 sec)

mysql> insert into EMP values("E8", "J.Jones", ".Syst.Anal");
Query OK, 1 row affected (0.00 sec)

Steps-5: Divided the EMP table into engineer_EMP:
CREATE TABLE engineer_EMP AS
-> SELECT * FROM EMP

```
    -> WHERE Title LIKE '%Eng';
Query OK, 4 rows affected (0.04 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

Steps-6: Divided the EMP table into anal_EMP:

```
CREATE TABLE anal_EMP AS
    -> SELECT * FROM EMP
    -> WHERE Title LIKE '%Anal';
Query OK, 3 rows affected (0.04 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

Step-7: Retrieve all rows from the EMP table where the Title is alphabetically less than "Programmer."

```
 select * from EMP where Title < "Programmer";
```

Step-8: Retrieve all rows from the EMP table where the Title is alphabetically greater than "Programmer."

```
 Select * from EMP where Title > "Programmer";
```

Step-9: Retrieve all rows from the EMP table where the Title is alphabetically less than or equal to "Programmer."

```
Select * from EMP where Title <= "Programmer";
```

Step-10: Retrieve all rows from the EMP table where the Title is alphabetically greater than or equal to "Programmer."

```
Select * from EMP where Title >= "Programmer";
```

Step-11: Retrieve the Eno (Employee Number), Ename (Employee Name), and Title from the anal_emp table and combine it with the Eno, Ename, and Title from the engineer_emp table using a UNION (i.e., merging the results from both tables).

Select Eno, Ename, Title from anal_emp union select Eno, Ename, Title from engineer_emp;

5. Result:

EMP Table

Eno	Ename	Title
E1	J.Doe	Elect.Eng
E2	M.Smith	Syst.Anal
E3	A.Lee	Mech.Eng
E4	J.Miller	Programmer
E5	B.Casey	Syst.Anal
E6	L.Chu	Elect.Eng
E7	R.Davis	Mech.Eng
E8	J.Jones	.Syst.Anal

Engineer Table

Eno	Ename	Title
E1	J.Doe	Elect.Eng
E3	A.Lee	Mech.Eng
E6	L.Chu	Elect.Eng
E7	R.Davis	Mech.Eng

Analylist Table

Eno	Ename	Title
E2	M.Smith	Syst.Anal
E5	B.Casey	Syst.Anal
E8	J.Jones	Syst.Anal

3 rows in set (0.00 sec)

Retrieve all rows from the EMP table where the Title is alphabetically less than "Programmer."

Eno	Ename	Title
E1	J.Doe	Elect.Eng
E3	A.Lee	Mech.Eng
E6	L.Chu	Elect.Eng
E7	R.Davis	Mech.Eng

Retrieve all rows from the EMP table where the Title is alphabetically greater than "Programmer."

Eno	Ename	Title
E2	M.Smith	Syst.Anal
E5	B.Casey	Syst.Anal
E8	J.Jones	Syst.Anal

select * from EMP where Title <= "Programmer";

Eno	Ename	Title
E1	J.Doe	Elect.Eng
E3	A.Lee	Mech.Eng
E4	J.Miller	Programmer
E6	L.Chu	Elect.Eng
E7	R.Davis	Mech.Eng

mysql> select * from EMP where Title >= "Programmer";

Eno	Ename	Title
E2	M.Smith	Syst.Anal
E4	J.Miller	Programmer
E5	B.Casey	Syst.Anal
E8	J.Jones	Syst.Anal

mysql> select Eno, Ename, Title from anal_emp union select Eno, Ename, Title from engineer_emp;

Eno	Ename	Title
E2	M.Smith	Syst.Anal
E5	B.Casey	Syst.Anal
E8	J.Jones	.Syst.Anal
E1	J.Doe	Elect.Eng
E3	A.Lee	Mech.Eng
E6	L.Chu	Elect.Eng
E7	R.Davis	Mech.Eng

6. Learning outcomes (What I have learned):

a) Learned how to split a table into multiple tables based on specific conditions.
b) Learned how to retrieve records by comparing string values alphabetically using operators like <, >, <=, and >=.
c) Learned the use of the UNION operator to combine data from multiple tables.

Evaluation Grid (To be created as per the SOP and Assessment guidelines by the faculty):

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Worksheet		8 Marks
2.	Viva		10 Marks
3.	Simulation		12 Marks
	Total		30 Marks

Teacher Signature